Like the cartographer, the string theorist can now claim with guarded optimism that the spectrum of logically sound theories incorporating the essential discoveries of the past century—special and general relativity; quantum mechanics; gauge theories of the strong, weak, and electromagnetic forces; supersymmetry; extra dimensions of Kaluza and Klein—is fully mapped out by Figure 12.11.

The challenge to the string theorist—or perhaps we should say the M-theorist—is to show that some point on the theory map of Figure 12.11 actually describes our universe. To do this requires finding the full and exact equations whose solution will pick out this elusive point on the map, and then understanding the corresponding physics with sufficient precision to allow comparisons with experiment. As Witten has said, “Understanding what M-theory really is—the physics it embodies—would transform our understanding of nature at least as radically as occurred in any of the major scientific upheavals of the past.”15 This is the program for unification in the twenty-first century.